

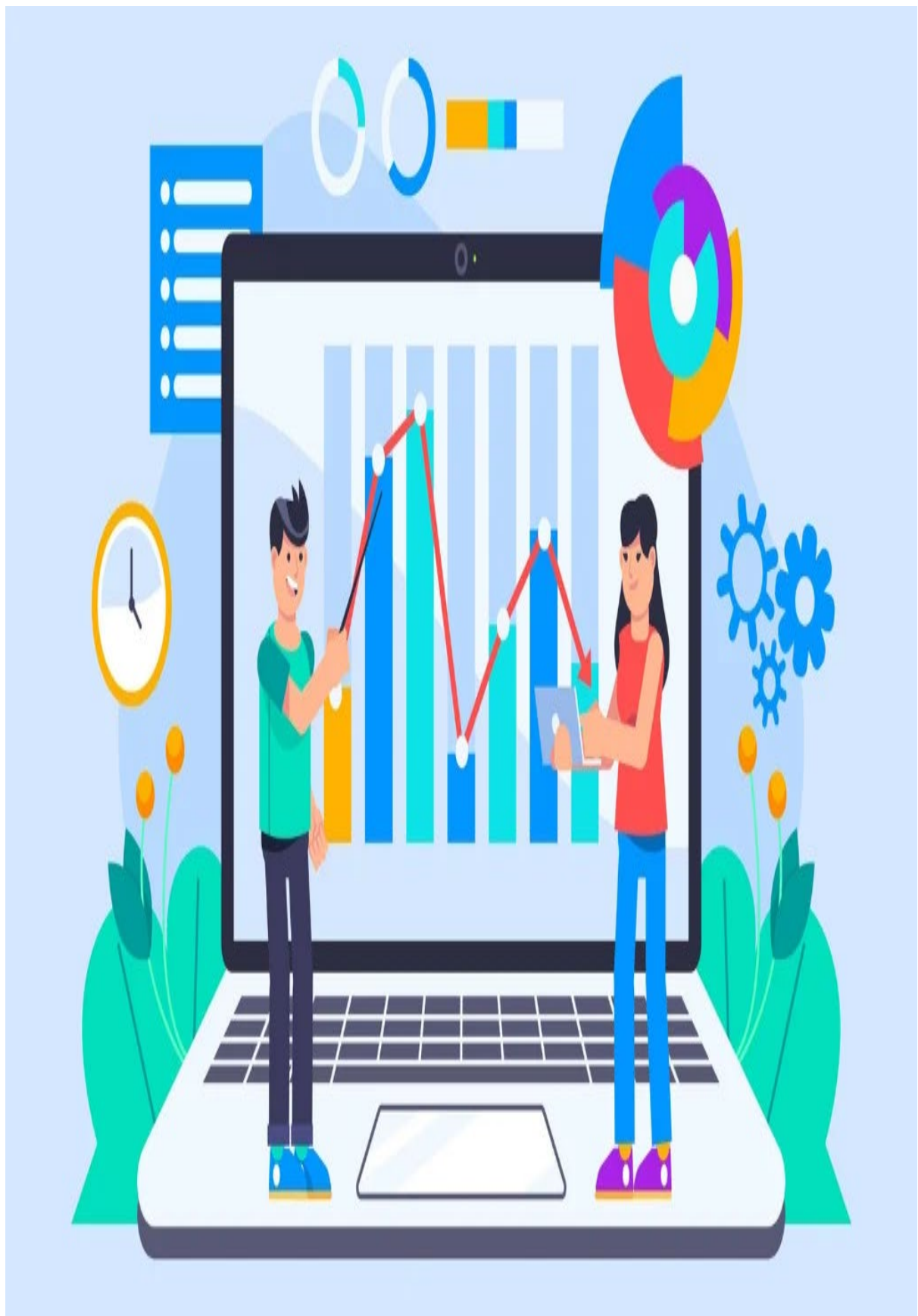
ABSTRACT

This project focuses on predicting student academic performance using machine learning regression models. Leveraging the Student Performance dataset, various features such as academic grades, personal background, and lifestyle choices were analyzed. After preprocessing and feature scaling, multiple algorithms including Linear Regression, Gradient Boosting, Ridge Regression, LightGBM, Random Forest, and CatBoost were evaluated. The performance of each model was measured primarily using the R^2 score, with the Linear Regression model achieving the highest score of 91.25%. This study highlights the potential of predictive analytics in identifying students at risk and guiding data-driven educational interventions.

INTRODUCTION

Predicting student academic performance is an important task in educational analytics, as it helps identify students who may need additional support. In this project, I used the Student Performance dataset, which includes a range of demographic, social, and academic features for secondary school students, such as age, family background, study habits, parental education, and previous grades.

The main objective was to build and evaluate machine learning models to predict students' final grades (G3). I focused on essential data preprocessing steps, including label encoding for categorical variables and MaxAbs scaling for numeric consistency, without performing additional feature engineering. By applying and comparing several regression algorithms, I aimed to assess how well these models can predict academic outcomes based solely on the original features in the dataset.



CONCLUSION

In this project, I built and evaluated several regression models to predict students' final grades using the Student Performance dataset. After encoding categorical features and scaling the data, I compared six algorithms: Linear Regression, Gradient Boosting, Ridge Regression, LightGBM, Random Forest, and CatBoost. Among these, Linear Regression achieved the highest R^2 score of 91.25%, indicating that the model explains over 91% of the variance in student performance.

This strong performance demonstrates the effectiveness of linear models in capturing the relationships within educational data, especially when the features are well-preprocessed and relevant. Moreover, the comparative analysis underscores the importance of evaluating multiple algorithms to identify the best fit for a given dataset. These findings suggest that predictive modeling can be a valuable asset for educators and policymakers, enabling proactive strategies to support at-risk students and improve academic outcomes through data-informed decision-making.